

Lecture 2: Kac Chaos, Propagation of Chaos, and the BBGKY Hierarchy

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Introduction

[CORE] The two main goals of this lecture are:

1. to introduce *Kac chaos* and *propagation of chaos* for stochastic exchangeable particle systems, and to clarify how these notions relate to (but differ from) mean-field convergence;
2. to develop the *BBGKY hierarchy* and to compare it with Dobrushin-type and entropy-based proof strategies, including the modern relative-entropy/modulated-energy program.

Recap of Lecture 1

[CORE] We proved Dobrushin’s stability estimate: under the assumption that the interaction kernel K is globally Lipschitz, two weak solutions $\rho_t, \tilde{\rho}_t$ of the mean-field continuity equation satisfy

$$W_1(\rho_t, \tilde{\rho}_t) \leq e^{2Lt} W_1(\rho_0, \tilde{\rho}_0).$$

Applying this with $\tilde{\rho}_t = \mu_t^N$ (the empirical measure of the particle system), we obtained the Dobrushin mean-field limit:

$$W_1(\mu_t^N, \rho_t) \leq e^{2Lt} W_1(\mu_0^N, \rho_0) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

DK:[Three things to keep in mind from Lecture 1: (i) the empirical measure μ_t^N solves the same nonlinear continuity equation as ρ_t , just with empirical initial data; (ii) Dobrushin’s bound says: mean-field convergence is just stability of the PDE in initial data; (iii) the proof was deterministic and used W_1 because the Lipschitz inequality pairs naturally with $|\cdot|^1$.]

Why go further?

[CORE] Dobrushin gives *convergence of the empirical measure*. But empirical-measure convergence does not directly give us:

- information about *individual* marginals $F_{N;j}$ for fixed j (e.g., the joint law of two tagged particles);
- strong distances such as relative entropy or total variation;
- access to singular kernels (Coulomb, Biot–Savart, Vlasov–Poisson, attractive Riesz);
- natural treatment of stochastic systems with Brownian noise (Lecture 1 outlook: Itô introduces fluctuations of size $N^{-1/2}$ that break the deterministic Dobrushin starting point).

These limitations motivate two complementary developments:

1. the *propagation-of-chaos* viewpoint, which tracks marginals rather than empirical measures;
2. the *BBGKY hierarchy*, which provides an exact (but unclosed) accounting of marginal dynamics, plus its modern counterparts (relative entropy, modulated energy).

1 Historical Motivation: Boltzmann to Kac and McKean

[CORE, can compress to 5 minutes]

1.1 The Stoßzahlansatz and the early kinetic theory of gases

The kinetic theory of gases originates with Clausius, Maxwell, and Boltzmann. The picture is a hard-sphere gas: many small balls moving freely in a box, occasionally colliding. The fundamental statistical assumption introduced by Boltzmann – the *Stoßzahlansatz* or molecular chaos hypothesis – is:

The velocities of two colliding particles are uncorrelated.

From this single hypothesis follow:

- the Maxwellian $M(v) = (2\pi T)^{-d/2} \exp(-|v - u|^2/2T)$ as the equilibrium distribution;
- the H-theorem: the entropy $H(f) = \int f \log f \, dv$ is non-increasing along the dynamics;
- the Boltzmann equation $\partial_t f + v \cdot \nabla_x f = Q(f, f)$.

DK:[Two reasons the Stoßzahlansatz cannot literally hold at the Newtonian level: (i) collisions create correlations – consecutive collisions are not independent; (ii) the Boltzmann equation is irreversible (H-theorem), while the underlying mechanics is time-reversible. Reconciling these tensions is the heart of the rigorous derivation problem.]

1.2 Hilbert’s 6th problem and the rigorous program

[CORE] Hilbert’s 6th problem (1900) asked, in essence, to develop mathematically the limiting processes that lead from the atomistic view to the laws of motion of continua:

- **Microscopic.** N identical particles in a state space E ; the configuration lives in E^N , an Nd -dimensional dynamical system
- **Mesoscopic** ($N \rightarrow \infty$). A typical particle has distribution $f_t \in \mathcal{P}(E)$; one studies the evolution of statistical quantities.

For deterministic hard spheres the rigorous theory is exceedingly delicate. The best results hold only in the dilute (Boltzmann–Grad) regime, for short times: Lanford (1975), extended by Gallagher–Saint-Raymond–Texier (2014). The Hilbert problem for hard spheres beyond short times remained open for decades.

DK:[Recent breakthrough [ADVANCED]: Deng–Hani (Chicago) and Hani–Ma (Michigan) have made dramatic progress on the long-time validity of the Boltzmann equation in low-density regimes. This is one of the most exciting recent developments in mathematical kinetic theory.]

1.3 Ehrenfest’s reframing and the mean-field route

[OPTIONAL] Tatyana and Paul Ehrenfest (1912) reframed the Stoßzahlansatz as a *statistical* assumption: factorization holds in a probabilistic sense for typical configurations, not as an exact mechanical statement.

The hard-sphere derivation being so hard, the field shifted toward *stochastic mean-field models*: point particles in a dense regime with continuous, rescaled interactions of size $1/N$.

Kac (1956): *Foundations of Kinetic Theory.* A stochastic mechanical caricature of the Boltzmann equation: random binary collisions at rate $O(1/N)$ per pair, with post-collisional velocities sampled from a fixed kernel. The N -particle process is a Markov jump process. Kac introduced the precise mathematical notion of *propagation of chaos*.

McKean (1967): *Propagation of chaos for a class of nonlinear parabolic equations.* N particles interact through a small smooth pairwise drift of size $1/N$, with independent Brownian noise per particle. The mean-field limit is a nonlinear Fokker–Planck equation.

DK:[The two prototypes generate the two main branches of stochastic propagation of chaos: jump processes (Boltzmann-type) and diffusions (Vlasov–Fokker–Planck-type). Kac uses semigroup expansion; McKean uses coupling. Both ideas survive in modern theory.]

Remark 1.1 (Boltzmann–Grad vs. mean-field). The classical derivation of the Boltzmann equation from *deterministic hard-sphere dynamics* is not a mean-field limit but the *Boltzmann–Grad limit*: $N \rightarrow \infty$ with the particle radius $r \rightarrow 0$ and Nr^{d-1} kept fixed (dilute regime). This is the setting of Lanford’s theorem (1975) and the recent breakthroughs of Deng–Hani and Hani–Ma. By contrast, when binary collisions are modeled as a stochastic Poisson jump process with rate λ/N , as in Kac’s caricature, the resulting limit *is* a mean-field limit and falls within the framework of this lecture. The two routes lead to (essentially) the same Boltzmann-type equation but via very different mathematics.

Remark 1.2 (Hard spheres versus mean-field jump models). For a hard-sphere gas, particles move freely until two spheres touch, at which point they undergo an elastic collision. The Boltzmann

equation is obtained in the Boltzmann–Grad scaling

$$N \rightarrow \infty, \quad \varepsilon \rightarrow 0, \quad N\varepsilon^{d-1} \sim 1,$$

which keeps the mean free path of order one. This is different from the Vlasov-type mean-field scaling: collisions are rare but strong, rather than weak and simultaneous.

In Kac or Nanbu models, by contrast, binary collisions are imposed stochastically through a jump generator. The Boltzmann collision operator is then the forward equation of a nonlinear mean-field jump process. The hard-sphere derivation is harder because the collision process must emerge from deterministic geometry, and one must control correlations and recollisions in order to justify molecular chaos.

2 The Framework: N -Particle Systems and Empirical Measures

[CORE] Throughout, E is a Polish state space (e.g. $\mathbb{R}^d$ or $\mathbb{R}^d \times \mathbb{R}^d$).

2.1 Markov generator setup

Definition 2.1 (N -particle system). A symmetric N -particle system is an E^N -valued Markov process $\mathcal{X}_t^N = (X_t^1, \dots, X_t^N)$ with law $f_t^N \in \mathcal{P}(E^N)$ characterized by the weak-forward Kolmogorov equation

$$\frac{d}{dt} \langle f_t^N, \varphi_N \rangle = \langle f_t^N, \mathcal{L}_N \varphi_N \rangle, \quad \varphi_N \in C_b(E^N),$$

where $\mathcal{L}_N$ is the Markov generator.

DK:[

Remark 2.2 (Generator viewpoint). The generator framework covers deterministic flows, diffusions, and jump processes. A nonlinear Markov process has a law-dependent generator $\mathcal{L}_{\mu_t}$, and its law satisfies the weak forward equation

$$\frac{d}{dt} \int \varphi d\mu_t = \int \mathcal{L}_{\mu_t} \varphi d\mu_t.$$

For a deterministic McKean–Vlasov flow,

$$\mathcal{L}_\mu \varphi(x) = b[\mu](x) \cdot \nabla \varphi(x).$$

For a McKean–Vlasov diffusion,

$$\mathcal{L}_\mu \varphi(x) = b[\mu](x) \cdot \nabla \varphi(x) + \frac{1}{2} a[\mu](x) : D^2 \varphi(x).$$

For a mean-field jump process,

$$\mathcal{L}_\mu \varphi(x) = \int [\varphi(y) - \varphi(x)] q[\mu](x, dy).$$

Thus the same forward-equation structure includes continuity equations, nonlinear Fokker–Planck equations, and Boltzmann-type collision equations.

]

Assumption 1 (Indistinguishability/Exchangeability). $\forall \pi \in \mathfrak{S}_N, (X_t^{\pi(1)}, \dots, X_t^{\pi(N)}) \sim (X_t^1, \dots, X_t^N)$.

DK:[Exchangeability is more than a labeling convenience. It is what allows the empirical measure to encode the full statistical content of the system: when particles are indistinguishable, all that matters is the unordered cloud, and the empirical measure captures exactly that.]

The empirical measure

$$\mu_{\mathcal{X}_t^N} := \frac{1}{N} \sum_{i=1}^N \delta_{X_t^i} \in \mathcal{P}(E)$$

is a *random* probability measure with law $F_t^N \in \mathcal{P}(\mathcal{P}(E))$.

Remark 2.3 (Three equivalent representations). For exchangeable systems, three objects encode the same information:

- the N -particle law $f_t^N \in \mathcal{P}(E^N)$;
- the random empirical measure $\mu_{\mathcal{X}_t^N}$, with law F_t^N ;
- the marginals $f_t^{s,N} \in \mathcal{P}(E^s)$ for $s \leq N$.

Each is convenient for different arguments. (Background on exchangeability and random measures: Dawson, St-Flour 1991.)

2.2 Two prototypes: Kac collision process

[CORE] Particles undergo random binary collisions; a Poisson process is associated with each pair $\{i, j\}$, with collision rate $\lambda(Z_t^i, Z_t^j)/N$. Hard-sphere example: $\lambda(x, y) = \delta_{|x-y|=2R}$.

When particles i, j collide, their post-collisional states are sampled from a kernel $\Gamma^{(2)}(z_1, z_2, dz'_1, dz'_2)$, symmetric in (z_1, z_2) .

Two-particle generator.

$$L^{(2)}\varphi_2(z_1, z_2) = \lambda(z_1, z_2) \int_{E^2} \{\varphi_2(z'_1, z'_2) - \varphi_2(z_1, z_2)\} \Gamma^{(2)}(z_1, z_2, dz'_1, dz'_2).$$

N -particle generator.

$$\mathcal{L}_N \varphi_N = \frac{1}{N} \sum_{i < j} L^{(2)} \diamond_{ij} \varphi_N,$$

where $L^{(2)} \diamond_{ij}$ acts on the (i, j) coordinates.

DK:[The $1/N$ factor is the mean-field scaling. With $\binom{N}{2} \sim N^2/2$ pairs and rate λ/N each, the total collision rate per particle is $O(1)$ as $N \rightarrow \infty$. Without the $1/N$ scaling, the system would freeze instantly.]

2.3 Two prototypes: McKean–Vlasov diffusion

[CORE] Now every particle feels a small force from every other, plus its own Brownian noise:

$$dX_t^i = F \star \mu_{\mathcal{X}_t^N}(X_t^i) dt + \sigma dB_t^i, \quad F \star \mu(x) := \int_E F(y - x) \mu(dy).$$

One-particle generator. For $\mu \in \mathcal{P}(E)$,

$$L_\mu \varphi(x) := F \star \mu(x) \cdot \nabla \varphi(x) + \frac{1}{2} \sigma^2 \Delta \varphi(x).$$

N -particle generator. $\mathcal{L}_N \varphi_N(\mathbf{x}^N) = \sum_{i=1}^N L_{\mu_{\mathbf{x}^N}} \diamond_i \varphi_N(\mathbf{x}^N)$.

Example (active particles). Self-propulsion + short-range repulsion:

$$\frac{dX_t^i}{dt} = V_t^i, \quad \frac{dV_t^i}{dt} = (1 - |V_t^i|^2) V_t^i - \frac{1}{N} \sum_{j=1}^N \nabla_{x^i} e^{-|X_t^j - X_t^i|/R}.$$

DK:[The McKean–Vlasov picture is the stochastic counterpart of last lecture’s deterministic mean-field continuity equation. Adding noise brings new tools (entropy, log-Sobolev) but breaks the exact-empirical-solution starting point of Dobrushin – because of the $O(N^{-1/2})$ Itô fluctuation term. We discussed this at the end of Lecture 1.]

3 Kac Chaos and Propagation of Chaos

[CORE]

3.1 Definitions

Definition 3.1 (Kac chaos at a fixed time). f_t^N is f_t -chaotic if for every fixed $s \in \mathbb{N}$,

$$f_t^{s,N} \xrightarrow[N \rightarrow \infty]{} f_t^{\otimes s} \quad \text{weakly in } \mathcal{P}(E^s).$$

The intuition: *when N is large, any fixed group of s tagged particles is approximately i.i.d. with law f_t .*

DK:[Critical phrase: “any FIXED group of s particles, with s kept fixed and $N \rightarrow \infty$.” If s scales with N , the situation is much more subtle (Lacker; Hess–Childs–Rowan). The case $s = N$ is the strongest; as it controls the entire N -particle law.]

Definition 3.2 (Propagation of chaos). The system propagates chaos if f_0^N being f_0 -chaotic implies f_t^N is f_t -chaotic for all $t \geq 0$.

Convention. Throughout, we assume the strongest initial condition $f_0^N = f_0^{\otimes N}$ (i.i.d. initial particles).

Lemma 3.3 (Kac’s lemma: equivalent formulations). *The following are equivalent for exchangeable (f_t^N) :*

1. Kac chaos for all s : $f_t^{s,N} \rightarrow f_t^{\otimes s}$ weakly in $\mathcal{P}(E^s)$ for every s ;
2. Kac chaos for the marginal $s = 2$ only;
3. the random empirical measure converges: $F_t^N \rightarrow \delta_{f_t}$ in $\mathcal{P}(\mathcal{P}(E))$.

Sketch of Kac’s lemma. The implications $(1) \Rightarrow (2)$ and $(1) \Rightarrow (3)$ are direct: marginal convergence for all s trivially gives it for $s = 2$, and convergence in law of $\mu_{\mathcal{X}_t^N}$ to δ_{f_t} as a random measure follows from convergence of all moments $\mathbf{E}\langle \mu_{\mathcal{X}_t^N}, \varphi_1 \rangle \cdots \langle \mu_{\mathcal{X}_t^N}, \varphi_s \rangle$ which equal $\langle f_t^{s,N}, \varphi_1 \otimes \cdots \otimes \varphi_s \rangle$ up to $O(1/N)$ corrections from coincident indices.

The non-trivial implication is $(2) \Rightarrow (1)$: two-particle chaos implies s -particle chaos for all s . The key tool is the variance identity for empirical measures: for $\varphi \in C_b(E)$,

$$\mathbf{Var}(\langle \mu_{\mathcal{X}_t^N}, \varphi \rangle) = \frac{1}{N} \mathbf{Var}_{f_t^{1,N}}(\varphi) + \frac{N-1}{N} \langle f_t^{2,N} - (f_t^{1,N})^{\otimes 2}, \varphi \otimes \varphi \rangle.$$

The first term is $O(1/N)$. The second term goes to zero by the assumed two-particle chaos $f_t^{2,N} \rightarrow f_t^{\otimes 2}$ together with $f_t^{1,N} \rightarrow f_t$ (which follows by integrating out one variable). Hence $\langle \mu_{\mathcal{X}_t^N}, \varphi \rangle \rightarrow \langle f_t, \varphi \rangle$ in L^2 , which gives $\mu_{\mathcal{X}_t^N} \rightarrow f_t$ in law as a random measure (i.e. $F_t^N \rightarrow \delta_{f_t}$). By continuity of the map $\mu \mapsto \mu^{\otimes s}$ on $\mathcal{P}(E)$, this implies $f_t^{s,N} \rightarrow f_t^{\otimes s}$ for every s . The full argument is combinatorial: see Sznitman (Saint-Flour 1989, Proposition I.2.2). $\square$

DK:[The equivalence “two-particle chaos $\Leftrightarrow$ all-orders chaos” is striking. Once you control the joint law of any two tagged particles, you automatically control the joint law of any fixed number s . see Sznitman St-Flour 1989.]

3.2 Mean-field limit vs. propagation of chaos

[CORE] These two notions are closely related but conceptually distinct. The cleanest way to see the relation is via Kac’s lemma above.

Mean-field limit (Lecture 1). Convergence of the empirical measure as a random variable (or, in the deterministic setting, as a deterministic curve):

$$\mu_{\mathcal{X}_t^N} \rightharpoonup f_t \quad (\text{weakly, in some sense}).$$

Propagation of chaos. Convergence of marginals: $f_t^{s,N} \rightarrow f_t^{\otimes s}$ for every fixed s .

Relation. By Kac’s lemma applied to the law of $\mu_{\mathcal{X}_t^N}$:

$$[F_t^N \rightarrow \delta_{f_t}] \iff [f_t^{2,N} \rightarrow f_t^{\otimes 2}] \iff \text{Kac chaos at time } t.$$

So at the level of *weak convergence*, mean-field convergence of the empirical measure to a deterministic limit $\Leftrightarrow$ Kac chaos. The two notions are, modulo language, the same statement at this level.

Where they differ. The notions diverge once one asks quantitative questions:

- Mean-field limit results typically give a *rate* for $W_p(\mu_{\mathcal{X}_t^N}, f_t)$ – a *global* object on $\mathcal{P}(E)$.
- Propagation of chaos can be quantified at the marginal level: $\|f_t^{k,N} - f_t^{\otimes k}\|_{TV}, H(f_t^{k,N} | f_t^{\otimes k})$, or even DK:[L^2 marginal distances]. These see information local to a fixed k .
- Sharp $O(k/N)$ marginal rates (Lacker 2021, Hess–Childs–Rowan) are statements that have no clean empirical-measure analogue.

DK:[“Mean-field limit and Kac chaos are the same statement at the level of weak convergence; they differ at the level of quantitative metrics. The marginal viewpoint is finer in the diffusive/quantitative setting.”]

3.3 Hierarchy of chaos: Kac, entropy, Fisher information

[CORE] Several non-equivalent levels of chaos are useful:

Definition 3.4 (Entropy chaos). (f^N) is *entropy-chaotic* with limit f if it is f -chaotic and $\frac{1}{N}H(f^N | f^{\otimes N}) \rightarrow 0$.

Definition 3.5 (Fisher-information chaos). (f^N) is *Fisher-info chaotic* with limit f if it is entropy-chaotic and $\frac{1}{N}I(f^N | f^{\otimes N}) \rightarrow 0$ (relative Fisher information).

Key message. There is a hierarchy of strength:

$$\text{Fisher-information chaos} \implies \text{entropy chaos} \implies \text{Kac chaos}.$$

The stronger notions become natural when diffusion, regularization, or quantitative information-theoretic control is desired.

DK:[For a first pass, Kac chaos is the most accessible. Entropy and Fisher chaos become essential when you have diffusion ($\sigma > 0$) and want to use functional inequalities (log-Sobolev): this is the modern Jabin–Wang program.]

Why entropy is useful (preview). The chain

$$\frac{1}{2}\|f^{k,N} - f^{\otimes k}\|_{TV}^2 \leq H(f^{k,N} | f^{\otimes k}) \leq \frac{k}{N}H(f^N | f^{\otimes N})$$

(Csiszár–Kullback–Pinsker + sub-additivity, Ben Arous–Zeitouni 1999) means: *controlling the global N -particle relative entropy automatically controls all marginals up to order k in TV , with explicit rate k/N* . This is the fundamental reason entropy methods are powerful. We return to this in Section 6.

4 The BBGKY Hierarchy

[CORE] BBGKY = Bogoliubov–Born–Green–Kirkwood–Yvon. It is the system of evolution equations satisfied by the marginals of an N -particle Liouville or Fokker–Planck equation.

DK:[Frame for students: “BBGKY is the natural language to think about marginals. We saw that propagation of chaos is a marginal statement. So we need an equation governing marginals – that is BBGKY. The catch is: the equation will be unclosed at every finite N .”]

4.1 Liouville equation and marginals

[CORE] For a deterministic first-order system

$$\dot{X}_i = -\frac{1}{N} \sum_{j=1}^N \nabla V(X_i - X_j), \quad i = 1, \dots, N,$$

with symmetric N -particle density $F_N(t, X_N)$, Liouville’s equation reads

$$\partial_t F_N + \sum_{i=1}^N \nabla_{x_i} \cdot (F_N b_i^N) = 0, \quad b_i^N(X_N) = -\frac{1}{N} \sum_{j=1}^N \nabla V(x_i - x_j).$$

For diffusions, the corresponding equation is the N -particle Fokker–Planck:

$$\partial_t F_N + \sum_{i=1}^N \nabla_{x_i} \cdot (F_N b_i^N) = \sigma \sum_{i=1}^N \Delta_{x_i} F_N.$$

Marginals. For $j \leq N$, define

$$F_{N:j}(t, x_1, \dots, x_j) := \int F_N(t, x_1, \dots, x_N) dx_{j+1} \cdots dx_N.$$

The BBGKY hierarchy is the system of equations satisfied by $(F_{N:j})_{1 \leq j \leq N}$.

4.2 The first BBGKY equation: a concrete prototype

[CORE] Take the deterministic gradient system above. Integrating the Liouville equation in $x_2, \dots, x_N$:

$$\partial_t F_{N:1}(t, x_1) + \nabla_{x_1} \cdot \left[\frac{N-1}{N} \int_{\mathbb{R}^d} (-\nabla V(x_1 - x_2)) F_{N:2}(t, x_1, x_2) dx_2 \right] = 0.$$

DK:[Two things to highlight on the board: (i) the equation for $F_{N:1}$ is *driven by* $F_{N:2}$; so it is not closed; (ii) the factor $\frac{N-1}{N}$ encodes “a particle interacts with the other $N-1$, not with itself”; this is the leading $1/N$ correction that disappears in the limit.]

Proposition 4.1 (BBGKY hierarchy from Liouville). *Let $F_N(t, X^N)$ be a symmetric solution of the N -particle Fokker–Planck equation*

$$\partial_t F_N + \sum_{i=1}^N \nabla_{x_i} \cdot (F_N b_i^N) = \sigma \sum_{i=1}^N \Delta_{x_i} F_N, \quad b_i^N = \frac{1}{N} \sum_{j=1}^N K(x_i, x_j).$$

Then the j -particle marginal $F_{N:j}$ satisfies

$$\partial_t F_{N:j} = \sigma \sum_{i=1}^j \Delta_{x_i} F_{N:j} - \sum_{i=1}^j \nabla_{x_i} \cdot \left[F_{N:j} \cdot \frac{1}{N} \sum_{k=1}^j K(x_i, x_k) \right] - \sum_{i=1}^j \nabla_{x_i} \cdot \left[\frac{N-j}{N} \int K(x_i, y) F_{N:j+1}(t, x_1, \dots, x_j, y) dy \right].$$

Sketch. Integrate the Liouville/Fokker–Planck equation against $dx_{j+1} \cdots dx_N$. The contributions split as follows:

1. Untagged variables ($i > j$). Both transport $\nabla_{x_i} \cdot (F_N b_i^N)$ and diffusion $\Delta_{x_i} F_N$ become divergences in integration variables. By the divergence theorem and sufficient decay at infinity, these contribute zero.

2. Tagged variables ($i \leq j$). The gradient ∇_{x_i} commutes with integration in $x_{j+1}, \dots, x_N$ (since these are different variables). The drift $b_i^N = \frac{1}{N} \sum_k K(x_i, x_k)$ splits into:

- *Intra-group* ($k \leq j$): independent of integration variables, gives $\frac{1}{N} \sum_{k \leq j} K(x_i, x_k) \cdot F_{N:j}$.
- *Coupling* ($k > j$): by symmetry of F_N , all $(N-j)$ contributions are equal, so reduce to $\frac{N-j}{N} \int K(x_i, y) F_{N:j+1}(t, x_1, \dots, x_j, y) dy$.

3. Diffusion. $\Delta_{x_i} F_{N:j}$ for $i \leq j$ commutes with the integration.

Combining gives the stated form. The key inputs are the divergence theorem (boundary terms vanish), commutativity of partial derivatives with integration in different variables, and full symmetry of F_N (used to reduce $(N-j)$ identical untagged contributions to a single one). $\square$

4.3 General schematic form

[CORE] For pairwise interacting diffusions, the BBGKY hierarchy takes the schematic form

$$\partial_t F_{N:j} = \mathcal{L}_j F_{N:j} + \frac{N-j}{N} \mathcal{C}_{j+1} F_{N:j+1} + \frac{1}{N} \mathcal{R}_j F_{N:j},$$

where:

- $\mathcal{L}_j$ is the *closed j -particle generator* – the part of the dynamics that depends only on the first j coordinates. It contains the free transport / diffusion terms and the internal interactions among particles $\{1, \dots, j\}$ *rescaled by $1/N$* ;
- $\mathcal{C}_{j+1}$ is the *coupling to the $(j+1)$ -marginal*: it accounts for the average interaction of one of the first j particles with a typical particle outside the group;
- $\mathcal{R}_j$ is a *correction* of order $1/N$ coming from internal interactions among the first j particles – this vanishes in the limit $N \rightarrow \infty$.

DK:[

Explicit form (McKean–Vlasov diffusion). For the system $dX_t^i = b_0(X_t^i) dt + \frac{1}{N} \sum_j K(X_t^i, X_t^j) dt + \sqrt{2\sigma} dB_t^i$, the BBGKY hierarchy reads

$$\begin{aligned} \partial_t F_{N:j} = & \underbrace{\sum_{i=1}^j \nabla_{x_i} \cdot [b_0(x_i) F_{N:j}] + \sigma \sum_{i=1}^j \Delta_{x_i} F_{N:j}}_{\mathcal{L}_j F_{N:j} \text{ (free } j\text{-particle dynamics)}} \\ & - \underbrace{\frac{1}{N} \sum_{1 \leq i, k \leq j} \nabla_{x_i} \cdot [K(x_i, x_k) F_{N:j}]}_{(1/N) \mathcal{R}_j F_{N:j} \text{ (intra-group)}} - \underbrace{\frac{N-j}{N} \sum_{i=1}^j \nabla_{x_i} \cdot \int K(x_i, y) F_{N:j+1} dy}_{((N-j)/N) \mathcal{C}_{j+1} F_{N:j+1} \text{ (coupling)}}. \end{aligned}$$

For the deterministic mean-field setting (Golse, no b_0 , no σ), the free-dynamics term $\mathcal{L}_j$ vanishes and only the interaction terms remain.]

Explicit form for the gradient diffusion. For the diffusion system $dX_i = -\frac{1}{N} \sum_j \nabla V(X_i - X_j) dt + \sqrt{2\sigma} dB_i$, one obtains

$$\partial_t F_{N:j} = \sigma \sum_{i=1}^j \Delta_{x_i} F_{N:j} - \frac{N-j}{N} \sum_{i=1}^j \nabla_{x_i} \cdot \int (-\nabla V(x_i - x_{j+1})) F_{N:j+1} dx_{j+1} - \frac{1}{N} \sum_{i, k \leq j} \nabla_{x_i} \cdot (\nabla V(x_i - x_k) F_{N:j}).$$

The three terms correspond exactly to $\mathcal{L}_j$, $\mathcal{C}_{j+1}$, and $\mathcal{R}_j$ above.

Key message. BBGKY is the *exact* marginal dynamics. It is not a limit theorem on its own. The hierarchy is not closed because the j -equation depends on the $(j+1)$ -equation.

DK:[It's worth emphasizing: At every finite N , knowing $F_{N:j}$ requires knowing $F_{N:j+1}$, which requires $F_{N:j+2}$, and so on. The chain only closes at $j = N$, but then we're back at the full Liouville equation in Nd dimensions; no progress. The whole point of BBGKY is what happens in the limit.]

4.4 Infinite hierarchy and factorized solutions

[CORE] Send $N \rightarrow \infty$ with j fixed. The $\frac{1}{N}$ correction $\mathcal{R}_j$ vanishes, $\frac{N-j}{N} \rightarrow 1$, and one gets a formal *infinite hierarchy*

$$\partial_t F_{\infty:j} = \mathcal{L}_j^\infty F_{\infty:j} + \mathcal{C}_{j+1} F_{\infty:j+1}, \quad j = 1, 2, 3, \dots$$

This is a system of equations for an infinite family $(F_{\infty:j})_{j \geq 1}$.

The factorization ansatz. Try the tensorized solution $F_{\infty:j}(t) = f_t^{\otimes j}$. Substituting:

- $\mathcal{L}_j^\infty f_t^{\otimes j}$ reduces, by symmetry, to a sum of one-particle operators applied to one factor at a time;
- $\mathcal{C}_{j+1} f_t^{\otimes(j+1)}$ reduces to a self-interaction of f_t .

The result is: $F_{\infty:j} = f_t^{\otimes j}$ solves the infinite hierarchy if and only if f_t solves a single nonlinear PDE – the *mean-field equation*.

DK:[This is the conceptual punchline: *the mean-field PDE is precisely the closure condition that makes tensorized states solutions of the infinite hierarchy*. Tensorization at $t = 0$ propagates to tensorization at all t , provided the one-particle marginal evolves by the mean-field PDE.]

4.5 Factorization equivalence: a closer look

We saw that the factorized ansatz $F_{\infty:j} = f_t^{\otimes j}$ solves the infinite hierarchy if and only if f_t solves a single mean-field PDE. Let us verify this more carefully.

Reverse direction ($\Leftarrow$). Suppose f_t solves the mean-field PDE

$$\partial_t f_t = \sigma \Delta f_t - \nabla \cdot ((K * f_t) f_t).$$

Apply Leibniz rule to the tensor product:

$$\partial_t f_t^{\otimes j} = \sum_{i=1}^j f_t^{\otimes(i-1)} \otimes (\partial_t f_t) \otimes f_t^{\otimes(j-i)}.$$

On the other hand, by symmetry of the right-hand side of the infinite hierarchy,

$$\mathcal{L}_j^\infty(f_t^{\otimes j}) + \mathcal{C}_{j+1}^\infty(f_t^{\otimes(j+1)}) = \sum_{i=1}^j f_t^{\otimes(i-1)} \otimes [\sigma \Delta f_t - \nabla \cdot ((K * f_t) f_t)] \otimes f_t^{\otimes(j-i)}.$$

The two sides match exactly because each summand is the mean-field equation applied to a single factor. Hence the infinite hierarchy is satisfied.

Forward direction ($\Rightarrow$). Conversely, if $F_{\infty:j} = f_t^{\otimes j}$ solves the infinite hierarchy, then in particular the $j = 1$ equation reads

$$\partial_t f_t = \mathcal{L}_1^\infty f_t + \mathcal{C}_2^\infty(f_t \otimes f_t),$$

which is exactly the mean-field PDE. Higher j -equations are then automatically satisfied by the forward direction.

DK:[The key structural insight: tensor products evolve via Leibniz rule, which produces sum-of-single-particle terms; and the right-hand side of the infinite hierarchy decomposes into the same sum-of-single-particle form by symmetry. This matching is no accident : it is precisely the statement that mean-field is the correct closure of the infinite BBGKY.]

Why this matters for propagation of chaos. Suppose we have a sequence $F_{N:j}(t)$ with tensorized initial data $F_{N:j}(0) = f_0^{\otimes j}$. The Lanford strategy proceeds:

1. Establish uniform-in- N bounds (regularity, moments) on $F_{N:j}(t)$. By compactness, extract subsequential limits $F_{\infty:j}(t)$.
2. Show any limit point satisfies the infinite hierarchy (passing to the limit in the weak/distributional formulation).
3. Prove uniqueness for the infinite hierarchy. Since the tensor product $f_t^{\otimes j}$ (with f_t the unique mean-field PDE solution) is a solution of the hierarchy with the right initial data, uniqueness forces $F_{\infty:j} = f_t^{\otimes j}$.

The non-trivial steps are uniform bounds (1) and uniqueness (3). For smooth interactions, both are standard; for singular kernels, both become hard.

4.6 Lanford strategy: how BBGKY proves chaos

[CORE] A natural question: *the finite BBGKY is not closed, so how do we extract a limit theorem?*

The classical answer is the *Lanford strategy* (also known as the hierarchy method, or tightness–consistency–uniqueness):

1. **Tightness/compactness.** Prove that $(F_{N:j})_N$ is tight in an appropriate space. By Prokhorov, a subsequence converges: $F_{N:j} \rightarrow F_{\infty:j}$.
2. **Consistency.** Show that any limit point $(F_{\infty:j})_{j \geq 1}$ satisfies the infinite hierarchy. The $1/N$ corrections vanish in the limit.
3. **Uniqueness.** Prove that the infinite hierarchy has a unique solution for the given initial data.

If f_t solves the mean-field PDE, then $F_{\infty:j} = f_t^{\otimes j}$ is a solution of the infinite hierarchy. By uniqueness, *every* chaotic limit is the factorized one. $\Rightarrow$ propagation of chaos.

Key message. *Uniqueness for the infinite hierarchy implies propagation of chaos.* This is the core mechanism by which BBGKY proves chaos.

DK:[Important caveat: “uniqueness for the infinite hierarchy” is itself a hard problem. Lanford for hard spheres proves it for short times only. For McKean–Vlasov-type problems uniqueness usually follows from well-posedness of the mean-field PDE. The Lanford strategy is general and elegant, but in practice often delivers no rate.]

4.7 The “so what” of BBGKY: why use an unclosed system?

[OPTIONAL but useful] A natural objection: if BBGKY is unclosed for every finite N , what is it good for?

1. It is an exact accounting tool. The hierarchy is not an approximation; it is a lossless reformulation of the Liouville (or Fokker–Planck) equation. It projects a dN -dimensional dynamics onto a sequence of dj -dimensional dynamics. Even unclosed, it tells us *exactly* how the correlation between j particles is driven by the $(j + 1)$ -th.

2. It is the natural setup for the Lanford strategy. As above: tightness $\rightarrow$ consistency $\rightarrow$ uniqueness.

3. It makes the structure of corrections visible. The $1/N$ terms in the hierarchy encode *finite- N corrections*. Modern higher-order PoC results (Lacker, Hess–Childs–Rowan) extract them via expansions of the hierarchy.

4. It accommodates singular kernels. BBGKY plus compactness/entropy methods can be adapted to settings where Dobrushin’s Lipschitz argument fails: Coulomb, Biot–Savart, etc.

4.8 Where BBGKY sits in the proof-strategy landscape

[CORE] BBGKY is not a method on the same footing as Dobrushin or Sznitman; it is a *framework*. To turn it into a limit theorem one needs a closure ingredient. The methods we have seen split along two axes:

Method	Object tracked (Layer 1)	Closure / estimate (Layer 2)
Dobrushin	Empirical measure μ_t^N	W_1 stability via coupling + Gronwall
Sznitman	Pathwise X_t^N vs $\bar{X}_t$ (same BM)	L^2 pathwise + Itô isometry, gives W_2
Lanford-style	Marginals $F_{N:j}$ (BBGKY)	Tightness + uniqueness for the infinite hierarchy
Jabin–Wang	N -particle law F_N	Relative entropy + functional inequalities
Lacker	Marginals $F_{N:j}$ (BBGKY)	Marginal entropy hierarchy + log-Sobolev
Serfaty	N -particle law F_N or μ_t^N	Modulated energy / free energy

Table 1: Two-layer decomposition of mean-field methods.

DK:[The narrative: “BBGKY says *track marginals*. That’s a Layer-1 choice. What you do with them – Lanford uniqueness, marginal entropy estimates, L^2 control – is Layer 2. Different combinations give different methods. So BBGKY is a coordinate system, and the closure techniques are the analytic devices on that coordinate system.”]

Remark 4.2 (Beyond pairwise interactions). DK:[The BBGKY framework derived above for pairwise kernels $K(x_i, x_j)$ extends to systems with non-pairwise drift $V(\mu, x)$ that depends nonlinearly on the empirical measure – e.g. in mean-field control, mean-field games, or consensus-based optimization. Recent work of Arnese–Lacker (arXiv:2603.10988, 2026) establishes sharp $O(k^2/N^2)$ rate in relative entropy and uniform-in-time results in the strongly displacement convex regime, by adapting the BBGKY hierarchy method to this nonlinear-in-measure setting. The Hess–Childs–Rowan L^2 analysis (arXiv:2310.09654) and higher-order Bogolyubov corrections currently remain in the pairwise framework; extending them to non-pairwise settings, especially with multiplicative noise (as in CBO) or singular functionals, is an active research direction.]

5 McKean's Coupling Theorem

[CORE] We now prove a propagation-of-chaos theorem in the diffusive setting. The technique is the *synchronous coupling method*, which is conceptually different from the BBGKY/Lanford strategy above but achieves a related (and quantitatively stronger) conclusion.

5.1 Guessing the Limit Equation

[CORE, can compress] The first step in McKean's argument is to *guess* what equation the limit f_t should satisfy. This is a heuristic/formal derivation: we assume $\mu_{\mathcal{X}_t^N} \rightarrow f_t$ (in law as a random measure) and read off the equation f_t would have to satisfy. This step does *not* prove convergence; it merely identifies the candidate limit equation, against which the actual coupling argument will then be carried out.

Recall the McKean–Vlasov N -particle generator $\mathcal{L}_N \varphi_N(\mathbf{x}^N) = \sum_{i=1}^N L_{\mu_{\mathbf{x}^N}} \diamond_i \varphi_N(\mathbf{x}^N)$. Testing against a one-particle φ and using exchangeability:

$$\frac{d}{dt} \langle f_t^{1,N}, \varphi \rangle = \langle \mu_{\mathcal{X}_t^N}, L_{\mu_{\mathcal{X}_t^N}} \varphi \rangle.$$

If $\mu_{\mathcal{X}_t^N} \rightarrow f_t$ (in law), then formally $\partial_t f_t = L_{f_t}^* f_t$, i.e.

$$\partial_t f_t = -\nabla_x \cdot (b(x, f_t) f_t) + \frac{1}{2} \sum_{i,j=1}^d \partial_{x_i} \partial_{x_j} \{ \sigma_{ij} \sigma_{ij}^T(x, f_t) f_t \}.$$

This is the nonlinear Fokker–Planck equation.

DK:[Three reasons this is a “guess” rather than a proof: (i) We assume $\mu_t^N \rightarrow f_t$ in law – but proving this is precisely the goal of the theorem. (ii) The functional $\mu \mapsto \langle \mu, L_\mu \varphi \rangle$ is quadratic in μ , so the evolution of $f_t^{1,N} = \mu_t^N$ has a correlation correction term involving $f_t^{2,N} - (f_t^{1,N})^{\otimes 2}$. The closed nonlinear FP equation arises only *after* chaos is assumed – exactly the chicken-and-egg structure characteristic of mean-field limits. (iii) Multi-particle marginals are not pinned down by the one-particle evolution alone; chaos requires more information than guessing yields.

What this step *does* give us: a well-defined candidate equation (provided it is well-posed, see next theorem) and a target for the coupling – the i.i.d. nonlinear processes $\bar{X}_t^i$ are constructed to follow precisely this guessed limit, and the actual coupling argument proves convergence.]

5.2 Wasserstein-2 distance and the marginal-coupling bound

Definition 5.1 (W_2). For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, $W_2^2(\mu, \nu) := \inf_{X \sim \mu, Y \sim \nu} |X - Y|^2$.

If $\bar{X}_t^i \sim f_t$ are i.i.d., then for any coupling $(X^i, \bar{X}^i)$ with the right marginals,

$$W_2^2(f_t^{s,N}, f_t^{\otimes s}) \leq \sum_{i=1}^s |X_t^i - \bar{X}_t^i|^2 = s |X_t^1 - \bar{X}_t^1|^2 \quad (\text{by exchangeability}).$$

DK:[This bound is the entire reason why coupling is the right tool: *a single one-particle estimate* $|X_t^1 - \bar{X}_t^1|^2$ *controls all s -particle marginals simultaneously*. The art is choosing the right coupling.]

5.3 The synchronous coupling

[CORE] Consider the McKean–Vlasov N -particle system

$$dX_t^i = b(X_t^i, \mu_{\mathcal{X}_t^N}) dt + \sqrt{2} dB_t^i.$$

Construct N independent *nonlinear processes*, driven by the *same* Brownian motions and with the same initial conditions:

$$d\bar{X}_t^i = b(\bar{X}_t^i, f_t) dt + \sqrt{2} dB_t^i, \quad \bar{X}_0^i = X_0^i,$$

where $f_t = \text{Law}(\bar{X}_t^i)$ solves the nonlinear Fokker–Planck equation.

The $\bar{X}_t^i$ are i.i.d. – they are coupled to the X_t^i through the Brownian motions and initial conditions, but as a family among themselves they are independent.

DK:[Three features of the synchronous coupling worth saying out loud: (1) same Brownian motions $\Rightarrow$ stochastic terms cancel in the difference $X_t^i - \bar{X}_t^i$; (2) same initial conditions $\Rightarrow$ no error from initialization; (3) the $\bar{X}_t^i$ are i.i.d. thanks to the N independent BMs and i.i.d. initial data. This is the deterministic-style trick adapted to SDEs – and it is exactly the analog of Dobrushin’s coupling, but pathwise rather than measure-valued.]

5.4 Statement of McKean’s theorem

Theorem 5.2 (Well-posedness). *If b is bounded and Lipschitz with respect to W_2 ,*

$$|b(x, \mu) - b(y, \nu)| \leq C(|x - y| + W_2(\mu, \nu)),$$

then for every $T > 0$ the nonlinear Fokker–Planck equation has a unique solution $f \in C([0, T]; \mathcal{P}_2(\mathbb{R}^d))$, and the distribution-dependent SDE has a unique strong solution.

Theorem 5.3 (McKean). *Under the same Lipschitz hypothesis,*

$$\forall T > 0, \quad \lim_{N \rightarrow \infty} \left[\sup_{t \leq T} |X_t^i - \bar{X}_t^i|^2 \right] = 0.$$

In particular, by the marginal-coupling bound, $W_2^2(f_t^{s,N}, f_t^{\otimes s}) \rightarrow 0$ for every fixed s : propagation of chaos holds in W_2 .

5.5 Proof of McKean’s theorem

[CORE – the proof of the day] We follow Sznitman’s St-Flour exposition. C denotes a generic constant.

Step 1: SDE difference. Brownian terms cancel:

$$X_t^i - \bar{X}_t^i = \int_0^t \left[b(X_s^i, \mu_{\mathcal{X}_s^N}) - b(\bar{X}_s^i, f_s) \right] ds.$$

Step 2: Square and Cauchy-Schwarz.

$$\sup_{t \leq T} |X_t^i - \bar{X}_t^i|^2 \leq 2T \int_0^T |b(X_t^i, \mu_{\mathcal{X}_t^N}) - b(\bar{X}_t^i, f_t)|^2 dt.$$

Step 3: Insert an i.i.d. empirical measure. Let $\mu_{\bar{\mathcal{X}}_t^N} := \frac{1}{N} \sum_{j=1}^N \delta_{\bar{\mathcal{X}}_t^j}$. By triangle inequality and $(a + b)^2 \leq 2a^2 + 2b^2$:

$$\begin{aligned} |b(X_t^i, \mu_{\mathcal{X}_t^N}) - b(\bar{X}_t^i, f_t)|^2 &\leq 2 |b(X_t^i, \mu_{\mathcal{X}_t^N}) - b(\bar{X}_t^i, \mu_{\bar{\mathcal{X}}_t^N})|^2 \\ &\quad + 2 |b(\bar{X}_t^i, \mu_{\bar{\mathcal{X}}_t^N}) - b(\bar{X}_t^i, f_t)|^2. \end{aligned}$$

Step 4: Lipschitz controls the first term.

$$\begin{aligned} |b(X_t^i, \mu_{\mathcal{X}_t^N}) - b(\bar{X}_t^i, \mu_{\bar{\mathcal{X}}_t^N})|^2 &\leq C \left(|X_t^i - \bar{X}_t^i|^2 + W_2^2(\mu_{\mathcal{X}_t^N}, \mu_{\bar{\mathcal{X}}_t^N}) \right) \\ &\leq C \left(|X_t^i - \bar{X}_t^i|^2 + \frac{1}{N} \sum_{j=1}^N |X_t^j - \bar{X}_t^j|^2 \right) \\ &\leq 2C |X_t^i - \bar{X}_t^i|^2 \quad (\text{by exchangeability}). \end{aligned}$$

Step 5: LLN governs the second term.

$$|b(\bar{X}_t^i, \mu_{\bar{\mathcal{X}}_t^N}) - b(\bar{X}_t^i, f_t)|^2 \leq C W_2^2(\mu_{\bar{\mathcal{X}}_t^N}, f_t).$$

This is a pure law-of-large-numbers statement: the empirical measure of i.i.d. samples converges to the underlying law.

Step 6: Gronwall. Combining Steps 2–5:

$$\sup_{t \leq T} |X_t^i - \bar{X}_t^i|^2 \leq C_1 \int_0^T W_2^2(\mu_{\bar{\mathcal{X}}_t^N}, f_t) dt + C_2 \int_0^T |X_t^i - \bar{X}_t^i|^2 dt.$$

Gronwall:

$$\sup_{t \leq T} |X_t^i - \bar{X}_t^i|^2 \leq C_1 e^{C_2 T} \int_0^T W_2^2(\mu_{\bar{\mathcal{X}}_t^N}, f_t) dt.$$

Step 7: LLN closes the estimate. By the strong LLN for empirical measures of i.i.d. samples, $\mu_{\bar{\mathcal{X}}_t^N} \rightarrow f_t$ almost surely; uniform second moments + dominated convergence give $W_2^2(\mu_{\bar{\mathcal{X}}_t^N}, f_t) \rightarrow 0$. Hence the right-hand side vanishes. $\square$

DK:[Two ingredients are doing all the work: (i) the Lipschitz drift, which lets us close a Gronwall on the squared distance; (ii) the LLN for empirical measures of i.i.d. samples, which is the only place “information about N ” enters. Notice this is structurally identical to Dobrushin: pick a coupling, write the difference equation, use Lipschitz, close with Gronwall, plug in LLN at the initial point. The only differences are the metric (W_2 instead of W_1) and the location where the LLN is invoked.]

5.6 Quantitative rates

[OPTIONAL] The proof can be made quantitative. The size of $W_2^2(\mu_{\bar{\mathcal{X}}_t^N}, f_t)$ is governed by Fournier–Guillin (PTRF 2015):

$$W_2^2(\mu_{\bar{\mathcal{X}}_t^N}, f_t) \leq \begin{cases} O(N^{-2/d}) & \text{if } d > 4, \\ O(N^{-1/2} \log(1 + N)) & \text{if } d = 4, \\ O(N^{-1/2}) & \text{if } d < 4. \end{cases}$$

Sharper bound for convolution drifts. If $b(x, \mu) = F \star \mu(x)$ with F bounded Lipschitz, a direct CLT-type computation gives

$$|b(\bar{X}_t^i, \mu_{\bar{X}_t^N}) - b(\bar{X}_t^i, f_t)|^2 \leq \frac{C}{N},$$

yielding the dimension-free rate $|X_t^i - \bar{X}_t^i|^2 = O(1/N)$.

DK:[The dimension-dependent rate $N^{-2/d}$ for general drifts (vs. $1/N$ for convolution drifts) is the famous *curse of dimensionality* for empirical measures. It is fundamental: Wasserstein distance between an empirical measure and its limit is intrinsically slow in high dimensions. This is one motivation for the entropy methods we discuss next: relative entropy delivers dimension-free $1/N$ rates without going through W_2 .]

5.7 Variations and extensions

[OPTIONAL]

McKean’s original argument (1967). Predates Sznitman’s clean formulation. Synchronous coupling between an N -particle and an M -particle system; show the family is Cauchy in $L^2(\Omega; C([0, T]; \mathbb{R}^d))$; use Hewitt–Savage 0-1 law to conclude that the limit processes are independent.

Mean-field jump processes (Kac-type). Use Poisson-driven SDE representation; couple jumps via shared driving Poisson measures. References: Graham (1992), Andreis–Dai Pra–Fischer (2018) for L^1 frameworks; Cortez–Fontbona (2016, 2018), Fournier–Mischler (2016) for the Boltzmann–Kac case.

Pathwise vs. pointwise PoC. The bound $\sup_{t \leq T} |X_t^i - \bar{X}_t^i|^2 \rightarrow 0$ is naturally pathwise – it says that entire trajectories are asymptotically independent, not just their values at any fixed time. This is strictly stronger than pointwise PoC.

Martingale methods. An alternative to coupling: characterize the N -particle law via a martingale problem; prove tightness; identify limit points as solutions of the limit martingale problem; use uniqueness. This is essentially the Lanford strategy in path-space form. Tanaka (1983), Sznitman (1984), Oelschläger (1984), Gärtner (1988), Graham–Méléard (1997).

6 Entropy Methods and Modern Hierarchy Results

[CORE] The Lanford strategy and McKean’s coupling are powerful but have limitations:

- Lanford typically gives no rate.
- McKean’s coupling needs a Lipschitz drift – excludes Coulomb, Biot–Savart, Vlasov–Poisson kernels.

A complementary route, very active since 2016, is the *relative entropy method*.

DK:[Frame: “So far we’ve tracked μ_t^N (Dobrushin), pathwise particles (Sznitman), or marginals via tightness (Lanford). The entropy method tracks the whole N -particle law $F_N(t)$ via $\text{KL}(F_N \| f_t^{\otimes N})$. Diffusion is essential here – it provides extra coercivity through the Fisher-information dissipation.”]

6.1 Notation

[CORE]

- $D(\mu\|\nu) = \int \mu \log(\mu/\nu)$ is the Kullback–Leibler divergence.
- $\mathcal{H}_N(t) := \frac{1}{N} D(F_N(t)\|f_t^{\otimes N})$ is the *normalized* relative entropy, or entropy per particle.

6.2 Why entropy controls all marginals at once

[CORE]

Lemma 6.1 (Chaos from entropy bounds). *For any $k \leq N$, $F_N \in \mathcal{P}(E^N)$, $f \in \mathcal{P}(E)$:*

$$\frac{1}{2} \|F_{N:k} - f^{\otimes k}\|_{TV}^2 \leq D(F_{N:k}\|f^{\otimes k}) \leq \frac{k}{N} D(F_N\|f^{\otimes N}) = k \mathcal{H}_N.$$

The first inequality is Csiszár–Kullback–Pinsker. The second is *entropy sub-additivity* (Ben Arous–Zeitouni 1999; Ben Arous–Brunaud 1990).

DK:[The chain $\frac{1}{2} \|F_{N:k} - f^{\otimes k}\|_{TV}^2 \leq D(F_{N:k}\|f^{\otimes k}) \leq k \mathcal{H}_N$ is the workhorse: *control the global N -particle entropy and you automatically control all marginals up to order k , in TV , with rate k/N . This is the magic of relative entropy methods.*]

Key message. If $\mathcal{H}_N(t) = \frac{1}{N} D(F_N(t)\|f_t^{\otimes N}) \rightarrow 0$ as $N \rightarrow \infty$, then automatically $F_{N:k}(t) \rightarrow f_t^{\otimes k}$ in total variation, with rate k/N .

6.3 Bounding entropy via Girsanov

[OPTIONAL] For a McKean–Vlasov diffusion $dX_t^i = b(X_t^i, \mu_{\mathcal{X}_t^N}) dt + \sigma dB_t^i$, Girsanov’s theorem gives

$$H(f_I^N | f_I^{\otimes N}) = \frac{N}{2\sigma^2} \left[\int_0^T |b(X_t^1, \mu_{\mathcal{X}_t^N}) - b(X_t^1, f_t)|^2 dt \right].$$

This is an exact identity, not an inequality.

DK:[Conceptual point: Girsanov changes measure to make X_i^N look like an i.i.d. nonlinear process; the Radon–Nikodym derivative measures exactly how far the actual interaction is from the mean-field one. Squaring gives the entropy. Reference: Sznitman St-Flour 1989.]

6.4 The differential inequality and closure

[CORE] Unlike the BBGKY equations – which are coupled at the level of marginals – the global entropy $\mathcal{H}_N(t)$ typically satisfies a *self-contained* functional inequality.

For interacting diffusions $dX_i = b[\mu_t^N](X_i) dt + \sqrt{2\sigma} dB_i$,

$$\frac{d}{dt} \mathcal{H}_N(t) = -\sigma \mathcal{I}_N(t) + \mathcal{G}_N(t),$$

where:

- $\mathcal{I}_N(t) = \frac{1}{N} \int F_N |\nabla \log(F_N/f_t^{\otimes N})|^2 dx^N$ is the relative Fisher information (always nonnegative);
- $\mathcal{G}_N(t)$ is an “interaction error” coming from the difference between the true particle interaction and the mean-field interaction.

The key technical step is to show

$$\mathcal{G}_N(t) \leq \epsilon \sigma \mathcal{I}_N(t) + C_\epsilon \mathcal{H}_N(t) + O(N^{-\alpha}).$$

Then choosing ϵ small absorbs the most singular part of the interaction into the negative Fisher term, leaving

$$\frac{d}{dt} \mathcal{H}_N(t) \leq C \mathcal{H}_N(t) + O(N^{-\alpha}).$$

Gronwall closes the estimate.

Key message. While BBGKY tries to close the system in the limit $N \rightarrow \infty$ via hierarchy uniqueness, entropy production methods often close the estimate *at finite* N via functional inequalities. Diffusion provides the extra coercivity (Fisher information).

DK:[This is conceptually different from the coupling/hierarchy approach. Coupling: build two processes, bound their distance. Hierarchy: identify limit, prove uniqueness. Entropy: bound a global functional via a differential inequality, automatic by sub-additivity. Each has its niche.]

6.5 What entropy methods deliver

[OPTIONAL]

- Strengthens Wasserstein conclusions to TV (Malrieu 2001).
- For linear interactions $b(x, \mu) = K \star \mu(x)$:
 - Bounded $K \in L^\infty$: Jabin–Wang (J. Funct. Anal. 2016).
 - $K \in W^{-1,\infty}$ (rougher than bounded): Jabin–Wang (Invent. Math. 2018).
 - Singular gradient systems $K = -\nabla W$ including Coulomb-type: Bresch–Jabin–Wang (Duke MJ 2022), Serfaty (ICM 2018).
- No assumption on the form of b : Lacker (ECP 2018).
- Sharp $O(k/N)^2$ marginal rate: Lacker (2021), via marginal entropy hierarchy + log-Sobolev.

Modulated free energy / modulated entropy. A close relative, designed for gradient-flow or kinetic systems with energy-dissipation structure. Compares the N -particle system with the continuum limit through a relative-energy functional. Particularly powerful for Coulomb gases and Vlasov-type problems.

6.6 Two distinct entropy approaches

DK:[This whole subsection is a personal note] There are two qualitatively different entropy methods, which are sometimes conflated:

(a) Jabin–Wang: global N -particle entropy. Track $\mathcal{H}_N(t) = \frac{1}{N} H(F_N(t) | f_t^{\otimes N})$ directly. Derive a self-contained Gronwall inequality via Donsker–Varadhan variational formula and functional inequalities (log-Sobolev, large deviations). Marginal convergence follows by the sub-additivity bound $H(F_{N:k} | f^{\otimes k}) \leq k \mathcal{H}_N \leq (k/N) H(F_N | f^{\otimes N})$, giving rate $O(k/N)$. *BBGKY hierarchy is not used.*

References: Jabin–Wang JFA 2016, Inv. Math. 2018; Bresch–Jabin–Wang Duke MJ 2022.

(b) Lacker: marginal entropy hierarchy via BBGKY. Track $H_t^k = H(F_{N:k}(t) | f_t^{\otimes k})$ for each k separately. Use the BBGKY hierarchy for $F_{N:k}$ to derive a system of differential inequalities

$$\frac{d}{dt}H_t^k \leq c_1 \frac{k^2}{N^2} + c_2 k (H_t^{k+1} - H_t^k), \quad \frac{d}{dt}H_t^N \leq c_1,$$

and solve by induction. Yields the sharp marginal rate $O(k^2/N^2)$ – quadratically better than sub-additivity. *BBGKY hierarchy is essential.*

References: Lacker PMP 2023 (arXiv:2105.02983); Lacker–Le Flem PTRF 2023.

DK:[Conceptual difference: Jabin–Wang treats the N -particle distribution globally via large-deviation tools; Lacker exploits the internal coupling structure of BBGKY at the marginal level. Different mathematical mechanisms, complementary strengths.]

7 Comparison: BBGKY versus Dobrushin versus Entropy

[CORE]

7.1 Dobrushin route

Controls W_p on empirical measures. Elegant, efficient, gives explicit rates for Lipschitz interactions. Limited by the Lipschitz assumption and by the empirical-measure framing.

7.2 BBGKY/Lanford route

Follows exact marginal dynamics. Structural and natural for kinetic theory. Closure by tightness and uniqueness for the infinite hierarchy. Most general – handles deterministic and stochastic, jump and diffusion – but typically gives no rate.

7.3 Entropy route (Jabin–Wang, Lacker, Serfaty)

Tracks a global functional $\mathcal{H}_N(t)$ directly via a self-contained differential inequality. Gives sharp rates and handles singular kernels in the stochastic case. Requires diffusion for the Fisher-information dissipation.

7.4 Summary table

Method	Main object	Best for
Dobrushin / coupling	empirical measures, trajectories	Lipschitz, fast quantitative MFL
Sznitman syn- chronous coupling	pathwise particles vs. nonlinear process	stochastic Lipschitz drift
Lanford / hierarchy	marginals $F_{N:j}$	structural marginal analysis, jump systems
Jabin–Wang (relative entropy)	full law F_N	singular kernels, stochastic
Lacker (marginal en- tropy)	marginals + entropy	sharp $O(k/N)$ marginal rates
Serfaty (modulated energy)	modulated functional	Coulomb gases, energy-dissipation structures

Table 2: Summary of proof strategies for mean-field limits and propagation of chaos.

8 Outlook

[OPTIONAL] Three directions where the field is active.

8.1 Long-time behavior and uniform-in-time PoC

The estimates above are valid on a finite time horizon $[0, T]$. A natural question: are they uniform in T ? I.e., is there a bound

$$\sup_{t \geq 0} |X_t^i - \bar{X}_t^i|^2 \leq \frac{C}{N}?$$

Granular media as motivation. The McKean–Vlasov equation $\partial_t f_t = \nabla \cdot (f_t \nabla (V + W \star f_t)) + \Delta f_t$ describes the long-time relaxation of a confined interacting system.

Theorem 8.1 (Malrieu, SPA 2001). *If V is β -uniformly convex, W is symmetric and convex, and ∇W is locally Lipschitz with polynomial growth, then*

$$\sup_{t \geq 0} |X_t^i - \bar{X}_t^i|^2 \leq \frac{C}{N}.$$

Key idea. Itô on the squared distance gives $\frac{d}{dt} |X_t^i - \bar{X}_t^i|^2 \leq -2\beta |X_t^i - \bar{X}_t^i|^2 + \dots$, where $-2\beta |\cdot|^2$ is the Bakry–Émery coercive term that prevents Gronwall blow-up.

DK:[Confining potential = decay engine. Without it, paths drift apart and the coupling weakens; with it, they’re pushed back together. This is the central tension in long-time PoC.]

DK:[Also, note that CBO does not fit into this framework automatically since i. its drift is not gradient structure, (so can’t use convexity directly), and ii. non-constant diffusion]

8.2 Phase transitions: when uniform PoC fails

[OPTIONAL] The mean-field limit can have multiple invariant measures even when the particle system has a unique one. This obstructs uniform-in-time PoC.

Kuramoto example. $d\theta_t^i = \frac{\gamma}{N} \sum_j \sin(\theta_t^j - \theta_t^i) dt + dB_t^i$.

- $\gamma < 1$: unique invariant measure (uniform on S^1).
- $\gamma > 1$: family of synchronized (von Mises) invariant measures parametrized by phase $\theta_0 \in S^1$.

On time scales $O(N)$, the empirical phase θ_0 performs a Brownian motion on S^1 (Bertini–Giacomin–Poquet, PTRF 2014). PoC breaks down on this scale.

8.3 Singular kernels and recent breakthroughs

[ADVANCED, optional] Real models often have singular kernels:

- Cucker–Smale alignment with singular communication.
- Overdamped Keller–Segel and Coulomb-type interactions: $K(r) = \xi r/|r|^d$.
- Vlasov–Poisson, Biot–Savart, attractive Riesz.

Coupling-based responses. Cut-off and mollifier methods ($K_\varepsilon \rightarrow K$ with $\varepsilon = \varepsilon_N \rightarrow 0$). Carrillo–Choi–Salem (2019); Oelschläger (1985); Jourdain–Méléard (1998); Bolley–Cañizo–Carrillo (2011).

Entropy/modulated energy responses. Jabin–Wang (2018); Bresch–Jabin–Wang (2022); Serfaty (2018, 2020); Duerinckx (2016).

Boltzmann long-time validity (very recent). [ADVANCED] The Hilbert problem for hard spheres beyond Lanford’s short-time validity has seen breakthroughs by Deng–Hani (Chicago) and Hani–Ma (Michigan), establishing Boltzmann validity over much longer time scales in low-density regimes. This is one of the most striking recent developments in mathematical kinetic theory.

8.4 Applications

[OPTIONAL] Mean-field theory and propagation of chaos appear in:

- Particle methods for nonlinear PDEs (Bird DSMC 1970, vortex methods, Landau);
- Consensus-based / particle-swarm optimization (Totzeck 2021);
- MCMC sampling, including SVGD;
- Mean-field training of neural networks ($\mathcal{R}(\theta^N) = \sum_\ell \text{Loss}(Y^\ell, \frac{1}{N} \sum_i \sigma(X^\ell, \theta^i))$);
- Mean-field games and mean-field control.

DK:[For a math/physics-oriented audience, mention the ML connection only briefly; for an applied audience, lead with it. The point in either case: mean-field is a unifying language across surprisingly disparate fields.]

Reading map for further study

- L.-P. Chaintron and A. Diez, *Propagation of chaos: a review of models, methods and applications*, Kinet. Relat. Models 15 (2022), 895–1015 (Parts I and II). **The most comprehensive modern reference**, <https://arxiv.org/abs/2106.14812>
- A.-S. Sznitman, *Topics in propagation of chaos*, Lectures at École d’Été de Probabilités de Saint-Flour XIX (1989), Lecture Notes in Math. 1464, Springer 1991.
- S. Méléard, *Asymptotic behaviour of some interacting particle systems; McKean–Vlasov and Boltzmann models*, Lecture Notes in Math. 1627, Springer 1996.
- P.-E. Jabin and Z. Wang, *Mean field limit for stochastic particle systems*, Active Particles, Vol. 1, Birkhäuser 2017.
- P.-E. Jabin, *A review of the mean field limits for Vlasov equations*, Kinet. Relat. Models 7 (2014), 661–711.
- F. Golse, *On the dynamics of large particle systems in the mean field limit*, arXiv:1301.5494 (2016).
- Lacker, *Hierarchies, entropy, and quantitative propagation of chaos for mean-field diffusions*, arXiv:2105.02983 (2021); Lacker, ECP 23 (2018).
- Serfaty, *Mean field limit for Coulomb-type flows*, Duke MJ 169 (2020); ICM 2018.
- Hauray–Mischler, *On Kac’s chaos and related problems*, J. Funct. Anal. 266 (2014).